Fiscal Year:	FY 2016	**Task Last Updated:**	FY 01/14/2016
PI Name:	Wolverton, Scot Ph.D.		
Project Title:	Characterizing Plant Gravity Perception Systems		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	(1) Bioregenerative Life Support		
PI Email:	scwolver@owu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	740-368-3503
Organization Name:	Ohio Wesleyan University		
PI Address 1:	Botany and Microbiology Department		
PI Address 2:	61 S Sandusky St		
PI Web Page:			
City:	Delaware	**State:**	OH
Zip Code:	43015-2333	**Congressional District:**	12
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight (ILSRA) NNH14ZTT002N
Start Date:	03/16/2015	**End Date:**	03/15/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	6	**Monitoring Center:**	NASA KSC
Contact Monitor:	Levine, Howard	**Contact Phone:**	321-861-3502
Contact Email:	howard.g.levine@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX15AG55G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Plant growth is characterized by plasticity, allowing environmental cues like light and gravity to play a significant role in determining the overall form of the plant body. Statolith sedimentation in specialized cells represents one way that plants perceive gravity. Plants lacking functional statoliths show reduced but consistent gravity responses, suggesting the presence of a non-statolith sensory system. This proposal seeks to investigate the gravity perception machinery in roots of several genotypes of the higher plant Arabidopsis thaliana by taking advantage of the unique microgravity environment of Earth orbit on the International Space Station (ISS). One thrust of the work described here is to characterize the physiological properties of the putative non-statolith sensory system, including the threshold acceleration force required to activate it, and its capacity to modulate downstream growth regulation machinery. The second thrust is to determine whether the two gravity sensory systems control a single or multiple output systems.		

	Addressing this question requires the microgravity environment in order to quantify the kinetics of gravitropism in roots having or lacking functional statoliths under fractional g conditions. The overall outcome of the work proposed here will be an improved understanding of how higher plants perceive gravity and transduce this physical cue into directional growth, a fundamental requirement for improving plants for production on Earth and in long-term spaceflight missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The overall outcome of the work proposed here will be an improved understanding of how higher plants perceive gravity and transduce this physical cue into directional growth, a fundamental requirement for improving plants for production on Earth and in long-term spaceflight missions.
Task Progress:	Plants are extremely sensitive to gravity and other environmental cues, which they use to direct their growth. Even though scientists have been studying how plants sense and respond to gravity for over 100 years, we still do not fully understand these processes. One way plants detect gravity is through the falling of dense bodies inside of cells called statoliths. But plants that lack statoliths because of a genetic mutation can still partially detect gravity, suggesting there's at least one other way that plants can sense gravity. We plan to study these mutant plants in space, where there is almost no gravity, and add back small amounts of gravity by growing them on a centrifuge. This will allow us to more fully understand how the mutants are detecting and responding to gravity compared with normal plants. This year, which was our first year of funding for this project, we started to define and plan our spaceflight experiment. We identified a number of questions that need to be answered before our project can be scheduled for flight. We spent most of our time working on designing experiments to answer these questions. We confirmed that the plants we plan to use as our statolith mutants are the correct plants that truly lack statoliths and began a process of continuously producing fresh seeds for the flight experiment. We compared methods of seed surface sterilization to ensure that our seeds do not carry any contamination while also protecting the embryos from damage. We tested our mutants and normal seedlings under environmental conditions like they will experience on flight, including elevated carbon dioxide levels and various nutrient mixtures to identify the best way to conduct our experiments. We tested our seedlings for growth rates at various stages of development on the special membranes we will use to grow them on flight and compared this with seedling growth on agar, our usual method of growing seedlings. We confirmed that our seedlings will respond to the environmental cues we plan to use in the flight experiment to orient the growth of seedlings in the absence of gravity. We are finishing the last of our preparatory experiments now, in which we are measuring the germination rates of seeds stored in flight conditions for many months, so we can determine how long we can safely store our experiment before launch and while onboard the Space Station before our seeds begin to get too old to germinate and grow normally. In addition to these experiments to prepare for flight, we also started studying the gravity response pathway in a number of other mutants, to test whether some of these mutations might give insight into the non-statolith gravity sensing system. To perform these experiments, we use a special device that we made just for measuring gravity responses in plants and that lets us hold the plant very still in a specific position relative to gravity. We are carrying out these experiments in the lab on a continuous basis, looking at many different mutants for clues to gravity sensing in plants.
Bibliography Type:	Description: (Last Updated: 05/04/2026)
Abstracts for Journals and Proceedings	Waxman SE, Markus DM, Okoli OA, Tangbau DDH, Wolverton C. "Optimizing Growth Conditions for the Study of Plant Gravity Perception on the ISS." Presented at the 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. , Nov-2015